

This file contains the outputs we observed on the terminal on starting the 5G core network, the RAN Intelligent Controller (RIC), the gNodeB that is the base station and the User Equipment (UE).

The 5G Core network

```
etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x
pcfd: NOT_RUNNING
nssfd: NOT_RUNNING
bsfd: NOT_RUNNING
udrd: NOT_RUNNING
webui: NOT_RUNNING
etccomm@etccomm-Verit...:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed/5G_Core_Network$ ./run.sh
# Script: /home/etccomm/0-RAN-Testbed-Automation/5G_Core_Network/install_scripts/network_config.sh...
Adding TUN interface ogstun...
Adding TUN interface ogstun2...
Adding TUN interface ogstun3...
Checking and assigning IP addresses to TUN device...
Running patched Open5GS netconf.sh script...
net.ipv4.ip_forward = 1
net.ipv6.conf.all.forwarding = 1
Adding iptables MASQUERADE rule for IPv4...
Adding iptables MASQUERADE rule for IPv6...
Configured network settings for Open5GS.
Starting webui service...
Starting open5gs-mmed in background...
Starting open5gs-sgwcd in background...
Starting open5gs-smfd in background...
Starting open5gs-amfd in background...
Starting open5gs-sgwud in background...
Starting open5gs-upfd in background...
Starting open5gs-hssd in background...
Starting open5gs-pcrfd in background...
Starting open5gs-nrfd in background...
Starting open5gs-scpd in background...
Starting open5gs-ausfd in background...
Starting open5gs-udmd in background...
Starting open5gs-pcfd in background...
Starting open5gs-nssfd in background...
Starting open5gs-bsfd in background...
Starting open5gs-udrd in background...
mmed: RUNNING
sgwcd: RUNNING
smfd: RUNNING
amfd: RUNNING
sgwud: RUNNING
upfd: RUNNING
hssd: RUNNING
pcrfd: RUNNING
nrfd: RUNNING
scpd: RUNNING
ausfd: RUNNING
udmd: RUNNING
pcfd: RUNNING
nssfd: RUNNING
bsfd: RUNNING
udrd: RUNNING
webui: RUNNING
etccomm@etccomm-Verit...:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed/5G_Core_Network$
```

The RAN Intelligent Controller (RIC)

```
etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x
[NEAR-RIC]: Loading SM ID = 2 with def = ORAN-E2SM-KPM
[NEAR-RIC]: Loading SM ID = 146 with def = TC_STATS_V0
[iApp]: Initializing ...
[iApp]: nearRT-RIC IP Address = 127.0.0.1, PORT = 36422
[NEAR-RIC]: Initializing Task Manager with 2 threads
[E2AP]: E2 SETUP-REQUEST rx from PLMN 1.1 Node ID 3584 RAN type ngran_gNB
[NEAR-RIC]: Accepting RAN function ID 2 with def = ORAN-E2SM-KPM
[NEAR-RIC]: Accepting RAN function ID 3 with def = ORAN-E2SM-RC
[NEAR-RIC]: Accepting RAN function ID 142 with def = MAC_STATS_V0
[NEAR-RIC]: Accepting RAN function ID 143 with def = RLC_STATS_V0
[NEAR-RIC]: Accepting RAN function ID 144 with def = PDCP_STATS_V0
[NEAR-RIC]: Accepting RAN function ID 145 with def = SLICE_STATS_V0
[NEAR-RIC]: Accepting RAN function ID 146 with def = TC_STATS_V0
[NEAR-RIC]: Accepting RAN function ID 148 with def = GTP_STATS_V0
^C
[NEAR-RIC]: Abruptly ending with signal number = 2
etccomm@etccomm-Veritlon-Series:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed/RAN_Intelligent_Controllers/Flexible-RIC/flexric/build/examples/ric
$ ./nearRT-RIC
[UTIL]: Setting the config -c file to /usr/local/etc/flexric/flexric.conf
[UTIL]: Setting path -p for the shared libraries to /usr/local/lib/flexric/
[NEAR-RIC]: nearRT-RIC IP Address = 127.0.0.1, PORT = 36421
[NEAR-RIC]: Initializing
[NEAR-RIC]: Loading SM ID = 3 with def = ORAN-E2SM-RC
[NEAR-RIC]: Loading SM ID = 143 with def = RLC_STATS_V0
[NEAR-RIC]: Loading SM ID = 145 with def = SLICE_STATS_V0
[NEAR-RIC]: Loading SM ID = 144 with def = PDCP_STATS_V0
[NEAR-RIC]: Loading SM ID = 148 with def = GTP_STATS_V0
[NEAR-RIC]: Loading SM ID = 142 with def = MAC_STATS_V0
[NEAR-RIC]: Loading SM ID = 2 with def = ORAN-E2SM-KPM
[NEAR-RIC]: Loading SM ID = 146 with def = TC_STATS_V0
[iApp]: Initializing ...
[iApp]: nearRT-RIC IP Address = 127.0.0.1, PORT = 36422
[NEAR-RIC]: Initializing Task Manager with 2 threads
[E2AP]: E2 SETUP-REQUEST rx from PLMN 1.1 Node ID 3584 RAN type ngran_gNB
[NEAR-RIC]: Accepting RAN function ID 2 with def = ORAN-E2SM-KPM
[NEAR-RIC]: Accepting RAN function ID 3 with def = ORAN-E2SM-RC
[NEAR-RIC]: Accepting RAN function ID 142 with def = MAC_STATS_V0
[NEAR-RIC]: Accepting RAN function ID 143 with def = RLC_STATS_V0
[NEAR-RIC]: Accepting RAN function ID 144 with def = PDCP_STATS_V0
[NEAR-RIC]: Accepting RAN function ID 145 with def = SLICE_STATS_V0
[NEAR-RIC]: Accepting RAN function ID 146 with def = TC_STATS_V0
[NEAR-RIC]: Accepting RAN function ID 148 with def = GTP_STATS_V0
```

The gNodeB (Base Station)

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etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x
UE ae41: LCID 4: TX      15 RX      320 bytes

[NR_MAC] Frame.Slot 640.0
UE RNTI ae41 CU-UE-ID 1 in-sync PH 48 dB PCMAX 20 dBm, average RSRP -44 (16 meas)
UE ae41: UL-RI 1, TPMI 0
UE ae41: dlsch_rounds 52/0/0/0, dlsch_errors 0, pucch0_DTX 0, BLER 0.00164 MCS (0) 0
UE ae41: ulsch_rounds 799/0/0/0, ulsch_errors 0, ulsch_DTX 0, BLER 0.00000 MCS (0) 0 (Qm 2 deltaMCS 0 dB) NPRB 5 SNR 23.5 dB
UE ae41: MAC: TX      1527 RX      10672 bytes
UE ae41: LCID 1: TX      546 RX      346 bytes
UE ae41: LCID 2: TX      0 RX      0 bytes
UE ae41: LCID 4: TX      15 RX      320 bytes

[NR_MAC] Frame.Slot 768.0
UE RNTI ae41 CU-UE-ID 1 in-sync PH 48 dB PCMAX 20 dBm, average RSRP -44 (16 meas)
UE ae41: UL-RI 1, TPMI 0
UE ae41: dlsch_rounds 53/0/0/0, dlsch_errors 0, pucch0_DTX 0, BLER 0.00148 MCS (0) 0
UE ae41: ulsch_rounds 811/0/0/0, ulsch_errors 0, ulsch_DTX 0, BLER 0.00000 MCS (0) 0 (Qm 2 deltaMCS 0 dB) NPRB 5 SNR 23.5 dB
UE ae41: MAC: TX      1548 RX      10906 bytes
UE ae41: LCID 1: TX      546 RX      346 bytes
UE ae41: LCID 2: TX      0 RX      0 bytes
UE ae41: LCID 4: TX      15 RX      320 bytes

[NR_MAC] Frame.Slot 896.0
UE RNTI ae41 CU-UE-ID 1 in-sync PH 48 dB PCMAX 20 dBm, average RSRP -44 (16 meas)
UE ae41: UL-RI 1, TPMI 0
UE ae41: dlsch_rounds 54/0/0/0, dlsch_errors 0, pucch0_DTX 0, BLER 0.00133 MCS (0) 0
UE ae41: ulsch_rounds 824/0/0/0, ulsch_errors 0, ulsch_DTX 0, BLER 0.00000 MCS (0) 0 (Qm 2 deltaMCS 0 dB) NPRB 5 SNR 63.5 dB
UE ae41: MAC: TX      1569 RX      11158 bytes
UE ae41: LCID 1: TX      546 RX      346 bytes
UE ae41: LCID 2: TX      0 RX      0 bytes
UE ae41: LCID 4: TX      15 RX      320 bytes

[NR_MAC] Frame.Slot 0.0
UE RNTI ae41 CU-UE-ID 1 in-sync PH 48 dB PCMAX 20 dBm, average RSRP -44 (16 meas)
UE ae41: UL-RI 1, TPMI 0
UE ae41: dlsch_rounds 55/0/0/0, dlsch_errors 0, pucch0_DTX 0, BLER 0.00120 MCS (0) 0
UE ae41: ulsch_rounds 837/0/0/0, ulsch_errors 0, ulsch_DTX 0, BLER 0.00000 MCS (0) 0 (Qm 2 deltaMCS 0 dB) NPRB 5 SNR 23.5 dB
UE ae41: MAC: TX      1590 RX      11412 bytes
UE ae41: LCID 1: TX      546 RX      346 bytes
UE ae41: LCID 2: TX      0 RX      0 bytes
UE ae41: LCID 4: TX      15 RX      320 bytes

[NR_MAC] Frame.Slot 128.0
UE RNTI ae41 CU-UE-ID 1 in-sync PH 48 dB PCMAX 20 dBm, average RSRP -44 (16 meas)
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User Equipment (UE)

```
etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x etccomm@etccomm-Verit... x
[NR_MAC] UE 0 RNTI ae41 stats sfn: 768.8, cumulated bad DCI 0
DL harq: 53/0
UL harq: 813/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.8, nb symbols 7.0)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 896.8, cumulated bad DCI 0
DL harq: 54/0
UL harq: 825/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.8, nb symbols 7.1)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 0.8, cumulated bad DCI 0
DL harq: 55/0
UL harq: 838/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.2)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 128.8, cumulated bad DCI 0
DL harq: 57/0
UL harq: 851/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.3)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 256.8, cumulated bad DCI 0
DL harq: 58/0
UL harq: 864/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.4)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 384.8, cumulated bad DCI 0
DL harq: 59/0
UL harq: 877/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.4)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 512.8, cumulated bad DCI 0
DL harq: 61/0
UL harq: 889/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.5)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 640.8, cumulated bad DCI 0
DL harq: 62/0
UL harq: 915/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.8, nb symbols 7.5)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 768.8, cumulated bad DCI 0
DL harq: 64/0
UL harq: 973/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.3)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 896.8, cumulated bad DCI 0
DL harq: 65/0
UL harq: 986/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.4)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 0.8, cumulated bad DCI 0
DL harq: 67/0
UL harq: 998/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.5)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 128.8, cumulated bad DCI 0
DL harq: 68/0
UL harq: 1011/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.6)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 256.8, cumulated bad DCI 0
DL harq: 69/0
UL harq: 1024/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.6)
[NR_MAC] UE 0 RNTI ae41 stats sfn: 384.8, cumulated bad DCI 0
DL harq: 71/0
UL harq: 1037/0 avg code rate 0.2, avg bit/symbol 2.1, avg per TB: (nb RBs 5.7, nb symbols 7.7)
```

It is observed that the data frames are in synchronisation confirming that a successful connection has been established.